

Intro to Mixed Models

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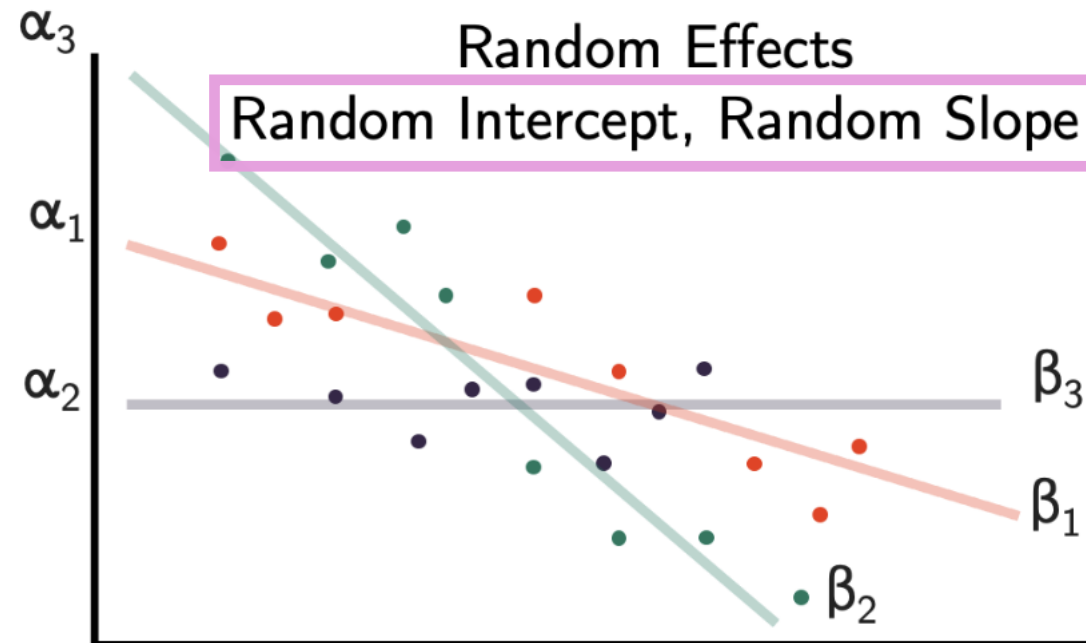
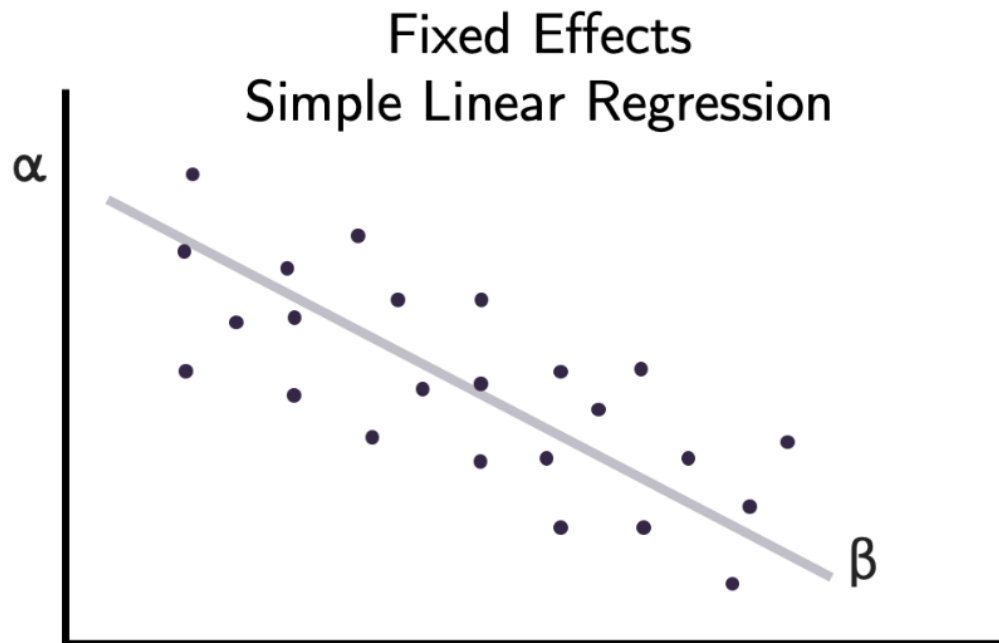
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Why do we need Mixed Models?

- Most standard statistical procedures (ANOVA, ordinary least-squares regression) assume data points are **independent and identically distributed (iid)**.
- In psychology, the independence assumption is almost always violated:
 - Repeated measures from the same participant.
 - Clustered/Nested data (students nested in classrooms, patients nested in hospitals).
- **The Consequence:** Ignoring non-independence underestimates standard errors, leading to heavily inflated Type I error rates (false positives).
- **The Solution:** Mixed Models

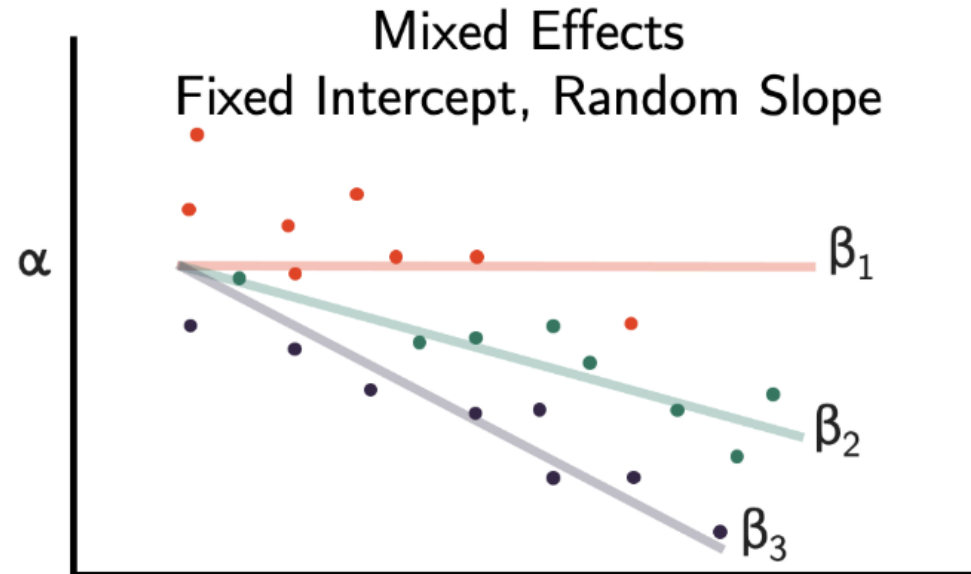
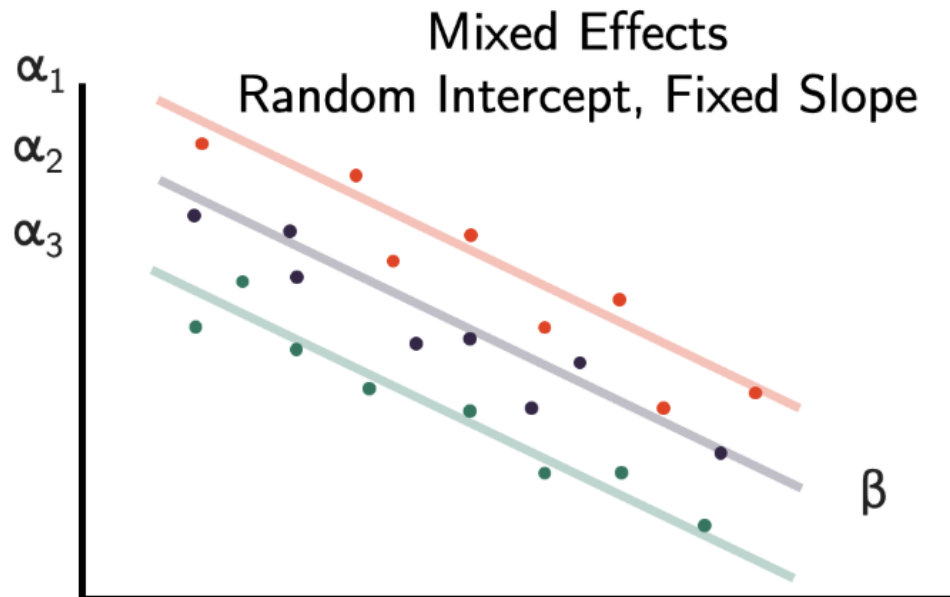
What is a Mixed Model?

- A **mixed model** includes:
 - Fixed effects → effects of interest (population-level)
 - Random effects → variability across groups (e.g., participants, items)



Random Intercepts vs. Random Slopes

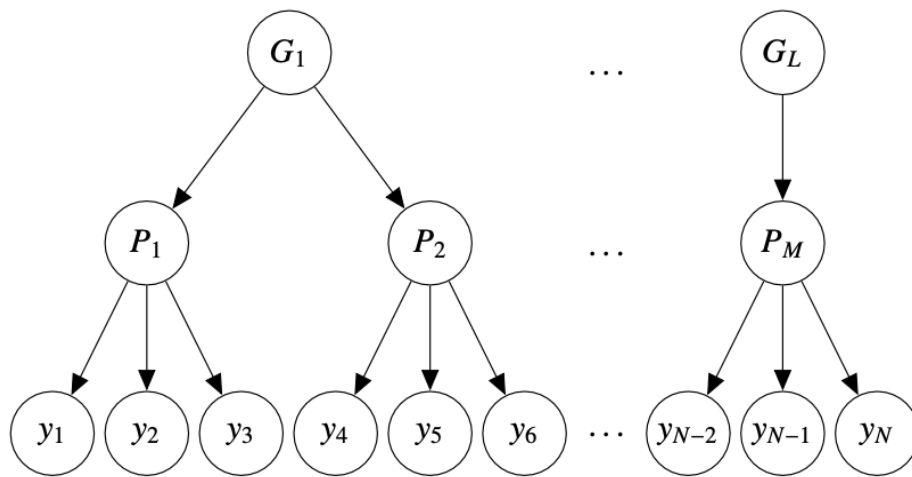
- Random Intercepts:
 - Assumes every group/participant has a different starting baseline.
- Random Slopes
 - Assumes the effect of the experimental condition is different for everyone.



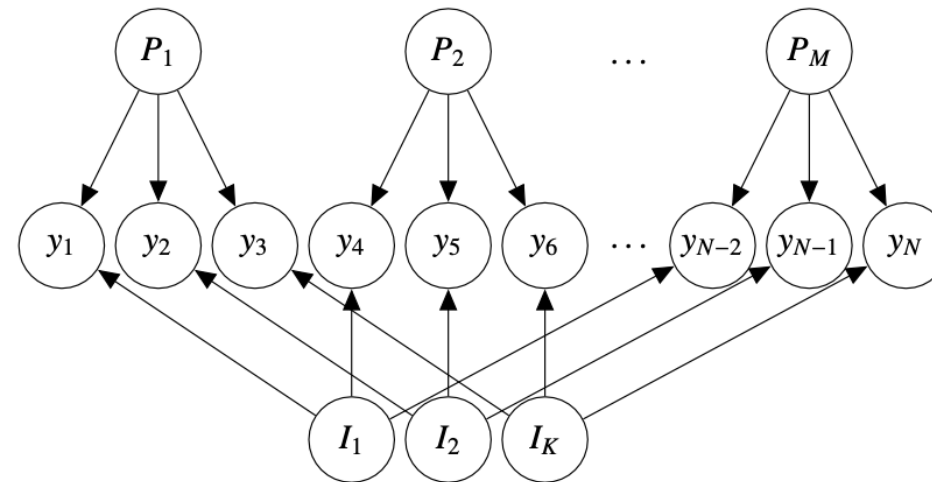
II Specifying Mixed Models

Types of Data Structure

- Nested (Hierarchical):
 - Lower-level units belong to exactly *one* higher-level unit.
 - Students nested within classrooms, or people nested within households.
- Crossed (Common in Experiments)
 - Observations belong to multiple, independent grouping factors simultaneously.
 - very participant responds to every item (word) in a task. Both "Participants" and "Items" are independent sources of random variation.



(c) Nested Random Effects



(b) Crossed Random Effects

II Specifying mixed models

Building Mixed Models

- Fixed-Effects-Only Model

$$\begin{aligned} y_{i,j,k} &= \beta_0 + \beta_\delta X_{i,j,k} + \epsilon_{i,j,k}, \\ i &= 1, 2, \dots, I, \quad j = 1, 2, \dots, J, \quad k = 1, 2, \\ \epsilon &\sim \mathcal{N}(0, \sigma_\epsilon^2), \end{aligned} \quad (1)$$

- Random Intercepts Model

$$\begin{aligned} y_{i,j,k} &= \beta_0 + S_{0,i} + \beta_\delta X_{i,j,k} + \epsilon_{i,j,k}, \\ i &= 1, 2, \dots, I, \quad j = 1, 2, \dots, J, \quad k = 1, 2, \\ \epsilon &\sim \mathcal{N}(0, \sigma_\epsilon^2), \\ S_0 &\sim \mathcal{N}(0, \sigma_{S_0}^2), \end{aligned} \quad (2)$$

$$RT \sim Condition + (1|Participant)$$

- Random-Intercepts-and-Slopes Model

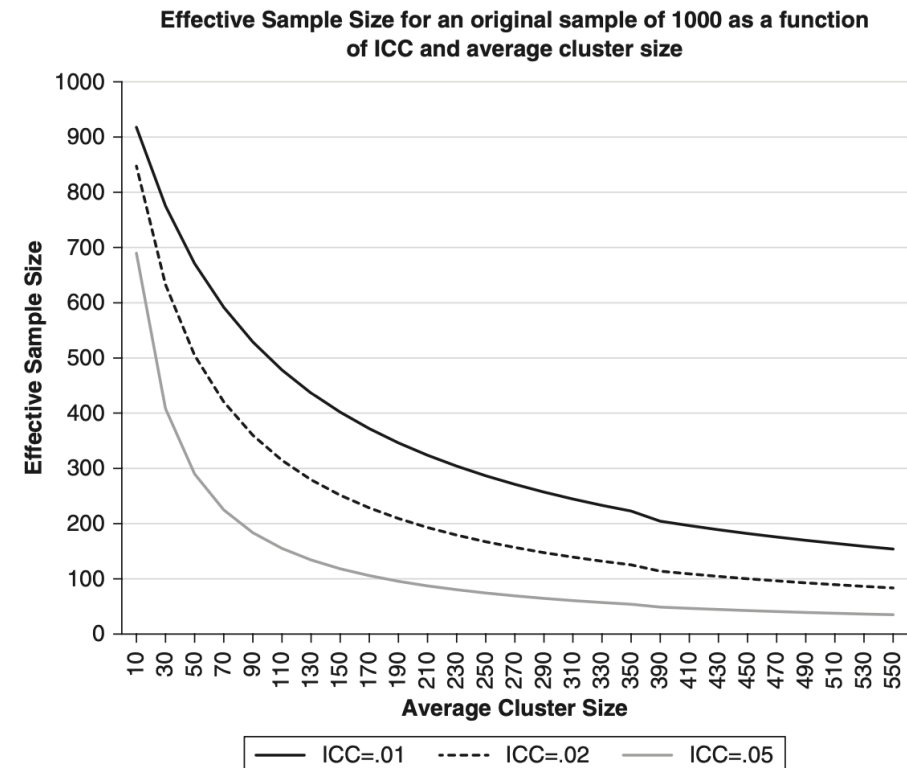
$$\begin{aligned} y_{i,j,k} &= \beta_0 + S_{0,i} + (\beta_\delta + S_{\delta,i})X_{i,j,k} + \epsilon_{i,j,k}, \\ i &= 1, 2, \dots, I, \quad j = 1, 2, \dots, J, \quad k = 1, 2, \\ \epsilon &\sim \mathcal{N}(0, \sigma_\epsilon^2), \\ \begin{pmatrix} S_0 \\ S_\delta \end{pmatrix} &\sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{S_0}^2 & \rho_{S_0, S_\delta} \sigma_{S_0} \sigma_{S_\delta} \\ \rho_{S_\delta, S_0} \sigma_{S_0} \sigma_{S_\delta} & \sigma_{S_\delta}^2 \end{bmatrix}\right), \end{aligned} \quad (3)$$

$$RT \sim Condition + (1 + Condition|Participant)$$

Why Clustering Matters: Intraclass Correlation Coefficient (ICC)

- **Intraclass Correlation Coefficient (ICC):**
 - Measures the proportion of total variability in the outcome that is explained by cluster membership.
 - Provides an estimate of the expected correlation between two randomly drawn individuals from the same cluster
- $ICC = 0 \rightarrow$ observations are independent
- $ICC > 0 \rightarrow$ observations within groups are correlated

$$\rho = \frac{\tau_{00}}{\tau_{00} + \sigma^2}$$



III Limitation and Discussion

Limitations of Mixed Models

- The Ideal (The Maximal Model):
 - To perfectly protect against Type I errors (false positives), we should include *all* random intercepts and random slopes justified by our design.
- In Real Applications
 - Complex mixed models require massive amounts of data.
 - Model Specification
 - Choosing the appropriate random-effects structure is not always straightforward
 - Interpretation
 - Requires careful explanation of fixed vs random effects

III Limitation and Discussion

Summary

- Psychological data are often **clustered / non-independent**
- Mixed models account for **variability across participants and items**
- Random intercepts and slopes capture **different sources of variation**
- Clustering reduces **effective sample size (ICC)**
- Ignoring it leads to **biased standard errors and false positives**
- Mixed models provide a **flexible framework but need careful interpretation**

Thank you!